



भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

Program Partner

CyberAI

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Secure your seat today to
lead the digital frontier

Course Starts on
01 Apr 2025

Course Duration
16 Weeks

Registration Ends
28 Feb 2025



Advanced Executive Program in **AI and Cybersecurity** by IIT Hyderabad

The Advanced Executive Program in AI and Cybersecurity by IIT Hyderabad is a premier educational offering designed for professionals aiming to excel in the rapidly evolving landscape of cybersecurity and artificial intelligence. This program blends academic rigor with practical industry insights, equipping participants with the knowledge and skills to address real-world challenges in the cybersecurity domain.

Program features

Why Choose This Program?

This program stands out for its focus on bridging the gap between theoretical foundations and practical applications. Participants will benefit from:



Practical Applications

Gain hands-on experience with real-world scenarios to build actionable expertise.



Case Studies

Learn through detailed case studies that provide insights into the challenges and solutions implemented by leading organizations.



Live Industry Applications

Engage with live applications from top industry experts, enabling you to stay ahead in this dynamic field



Comprehensive Curriculum

A meticulously curated syllabus that covers key areas of cybersecurity and AI, including threat detection, predictive analytics, and ethical hacking.



Guidance from Renowned Faculty

Learn from esteemed IIT Hyderabad faculty and industry leaders who bring a wealth of knowledge and expertise to the program.



Program Duration

The program spans **4 months**, providing a structured learning experience.



Delivery Mode

The course is delivered **online**, allowing flexible learning from anywhere.



Certification

Upon completion, you will receive a **certificate from IIT Hyderabad**, adding credibility and recognition to your skills.

The clock's ticking!. Join Us

Registration Ends on
28 FEB 2025

Course Starts on
01 APR 2025

Duration
16 Weeks
Online, 8-10 hours a week

Program fee
₹75,000 + GST
*Application Fee ₹1000

Course Offerings

Fueling Your Learning Journey

From basics to advanced hacks, here's what you'll master:

Phase 1

16 Hours

AI Fundamentals

- **Machine Learning Basics**
Understand ML types, training methods, feature engineering, metrics (2 Hrs)
- **Deep Learning Essentials**
Master neural networks, activation functions, loss functions, optimization (2 Hrs)
- **Advanced ML Concepts**
Learn ensemble methods, transfer learning, DL architectures, evaluation
- **AI Ethics and Bias**
Understand fairness, bias detection, responsible AI, transparency

Cybersecurity Fundamentals

- **Network Security**
Master TCP/IP, protocols, network architecture, traffic analysis
- **Security Controls**
Implement authentication, authorization, access control, policies
- **Threat Landscape**
Identify attack types, threat actors, vectors, risk assessment
- **Security Operations**
Execute incident response, log analysis, monitoring, threat intel

Phase 2

24 Hours

IDS

- **ML-based IDS Architecture**
Design ML-IDS architecture, implement detection systems
- **Real-time Detection Systems**
Build real-time detection, optimize performance
- **Hybrid Detection Approaches**
Combine multiple detection methods, enhance accuracy

Anomaly Detection

- **Statistical Methods**
Apply statistical anomaly detection methods

IoT Security

- **Edge Device Protection**
Secure IoT devices, implement edge protection
- **Lightweight ML Models**
Deploy efficient models, optimize for constraints
- **Distributed Detection**
Implement distributed security, coordinate detection

Behavior Analysis

- **User Behavior Analytics**
Profile user behavior, detect anomalies

Course Offerings

Fueling Your Learning Journey

From basics to advanced hacks, here's what you'll master:

Phase 2

24 Hours

Anomaly Detection

- **Deep Learning Approaches**
Implement DL-based anomaly detection
- **Autoencoder-based Detection**
Use autoencoders for anomaly detection
- **Real-time Implementation**
Deploy real-time anomaly detection systems

Biometric Authentication

- **Facial Recognition**
Implement facial recognition systems
- **Multimodal Biometrics**
Combine multiple biometric factors
- **Anti-spoofing Techniques**

Supply Chain Attacks

- **Software Supply Chain Attacks**
Software Supply chain attacks and how to protect against them
- **Performance Optimization**
Optimize system performance

Behavior Analysis

- **System Behavior Profiling**
Monitor system behavior, identify patterns
- **Network Behavior Analysis**
Analyze network behavior, detect threats

Vulnerability Scanning

- **ML-enhanced Scanning**
Improve scanning with ML techniques
- **Automated Assessment**
Automate vulnerability assessment
- **Risk Prioritization**



Course Offerings

Fueling Your Learning Journey

From basics to advanced hacks, here's what you'll master:

Phase 3

10 Hours

Adversarial Attacks

- **White-box Attacks**
Understand and implement white-box attacks
- **Black-box Attacks**
Master black-box attack techniques
- **Transfer Attacks**
Execute transfer attack methods
- **Physical Attacks**
Implement physical adversarial attacks

Model Extraction

- **API-based Extraction**
Extract models via API attacks
- **Side-channel Attacks**
Implement side-channel attacks
- **Membership Inference**
Execute membership inference attacks
- **Architecture Reconstruction**
Reconstruct model architectures

Data Poisoning

- **Training Data Poisoning**
Understand training data poisoning
- **Label Flipping**
Implement label flipping attacks
- **Backdoor Attacks**
Execute backdoor attack techniques
- **Clean Label Attacks**
Master clean label poisoning

Advanced Attacks

- **Model Inversion**
Implement model inversion attacks
- **Distillation Attacks**
Execute distillation attacks
- **Trojan Attacks**
Implement trojan attack methods
- **Privacy Leakage**
Understand privacy leakage attacks



Course Offerings

Fueling Your Learning Journey

From basics to advanced hacks, here's what you'll master:

Phase 4

20 Hours

Differential Privacy

- **DP Fundamentals**
Master DP concepts and mathematics
- **Privacy Budgeting**
Implement privacy budget management
- **DP-SGD Implementation**
Deploy DP-SGD in practice
- **Privacy Analysis**
Analyze privacy guarantees

Robust Defenses

- **Input Sanitization**
Implement input sanitization methods
- **Model Hardening**
Apply model hardening techniques
- **Certified Defenses**
Deploy certified defense methods
- **Detection Methods**
Implement attack detection systems

Federated Learning

- **FL Architecture**
Design FL systems architecture
- **Secure Aggregation**
Implement secure aggregation protocols
- **Cross-silo FL**
Deploy cross-silo FL systems
- **Privacy Guarantees**
Ensure FL privacy guarantees

Adversarial Training

- **PGD Training**
Execute PGD adversarial training
- **Ensemble Training**
Implement ensemble defenses
- **Verification Methods**
Apply verification techniques
- **Robustness Evaluation**
Evaluate model robustness



Benefits

Certificate Recognition

Elevate your career with the Advanced Executive Program in AI and Cybersecurity by IIT Hyderabad. Designed for professionals aiming to excel in the fields of cybersecurity and artificial intelligence, this program offers a comprehensive and hands-on learning experience. Participants will master the skills needed to tackle real-world challenges in network security, threat detection, and AI-driven cybersecurity solutions, paving the way for a rewarding and impactful career.

This program stands out for its focus on practical applications, live industry insights, and advanced tools, making it the ideal choice for professionals ready to advance their expertise.

Industry-Relevant Insights: Benefit from live industry applications and case studies guided by top experts and practitioners.

Prestigious Certification: Earn recognition with a certificate from IIT Hyderabad, enhancing your professional credentials and career prospect



Skill Covered

- Fundamentals of AI
- Core cybersecurity concepts
- Intrusion detection system development and optimization
- IoT security techniques
- Statistical and deep learning for anomaly detection
- Behavior analysis
- Biometric authentication
- Vulnerability scanning and assessment
- Adversarial attack countermeasures
- Data poisoning mitigation
- Model extraction prevention
- Advanced adversarial attack countermeasures
- Differential privacy techniques
- Federated learning
- Robust defense development
- Adversarial training techniques

About CyberAI

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OUR VISION

To lead the world in creating a secure digital future by merging Artificial Intelligence and Cybersecurity, empowering professionals to outpace tomorrow's challenges today.



OUR VALUES

- Purpose-Driven Innovation
- Empowerment Through Knowledge
- Integrity in Action
- Excellence for the Future
- Community & Collaboration



OUR MISSION

To equip ambitious minds with cutting-edge skills in AI and cybersecurity. Through hands-on learning, we empower learners to lead the charge in the digital frontier with resilience and innovation.

CyberAI¹

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